MARANDA 2025 KCSE PREDICTION

EXPECTED EXAM 1

ELECTRICITY

448/1

PAPER 1 (Theory)

TIME: 2½ HOURS

NAME……………………………………………………………………………………

SCHOOL………………………………………………… SIGN………………………

INDEX NO……………………………………………… ADM NO……………………

Kenya Certificate of Secondary Education.

INSTRUCTIONS TO CANDIDATES

*a) Write your **Name**, **Index number** and **Class** in spaces provided*
b) Candidates should have the following for this examination
Drawing instruments
Scientific calculator
*c) This paper consists of two sections: **A** and **B**.*
*d) Attempt **ALL** questions in **section A** and any **FOUR** in **section B**.*
*e) All dimensions are in **millimeters** unless stated other wise*

For Examiner's use only

Section	Questions	Maximum Score	Candidate's Score
A	1-10	48	
B	11	13	
	12	13	
	13	13	
	14	13	

	15	13	

SECTION A (48 marks)

(Attempt all questions in this section)

1. Explain the significance of using only one hand when working on high voltage system **(2 marks)**

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2. a) Define each of the following terms: **(3 marks)**

i. semiconductor

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ii. intrinsic semiconductor

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iii. extrinsic semiconductor

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b) Draw and label voltage versus current characteristics curve for a P-N junction in its forward and reverse modes **(3 marks)**

3. a) A 5kΩ linear potentiometer has 6V source connected across its end terminals

i. Determine the range of output voltage of the potentiometer **(1 mark)**

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ii. Calculate the two possible output voltages when the potentiometer is in its three-quarter position **(2 marks)**

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b) An emf of 120V causes a current of 3A to flow in a circuit. Calculate the energy supplied by the source in 12 seconds **(2 marks)**

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4. a) Outline two differences between a permanent magnet and an electromagnet **(2 marks)**

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b) State and explain two properties of a good conductor **(2 marks)**

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5. a) List four electrical measurements that can be taken using a cathode ray oscilloscope **(2 marks)**

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b) A moving coil meter gives a full scale deflection for a current of 100mA and has a resistance of 5Ω. Determine the value of the resistor used so that the meter can measure a current of 1.0A at full scale deflection. **(3 marks)**

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6. a) Draw the current discharge curve of a capacitor **(2 marks)**

b) Outline how insulation resistance test can be carried out between two conductors **(5 marks)**

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7. a) Name the logic gate shown in **Figure 1** (1 mark)

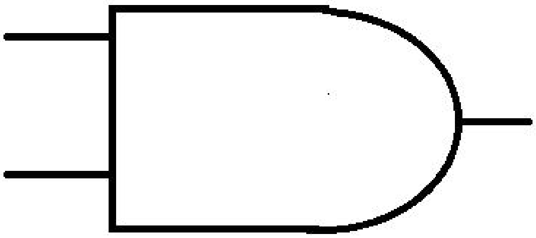

Figure 1

Name…………………………………………..

b) Complete the truth table of the OR gate **(2 marks)**

INPUT 1	INPUT 2	OUTPUT
0	0	
0	1	
1	0	
1	1	

c) Perform each of the following number conversions **(4 marks)**

i. 101010_2 to decimal

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ii. 23_{10} to binary

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8. a) Name four sources of electricity in Kenya **(2 marks)**

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b) State one possible cause and remedy for each of the following electric motor faults **(4 marks)**

i. Motor connected to the supply but will not start

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ii. Motor overheats on load

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9. Draw wiring diagram of a 12V electric bell controlled by two switches from a 240V mains supply **(4 marks)**

10. a) Define entrepreneur **(1 mark)**

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b) List two business opportunities in the field of electricity **(1 mark)**

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SECTION B (52 marks)

*Answer any **four** questions from this section in the spaces provided*

11. a) State three variables which determine resonance condition in an RLC circuit **(3 marks)**

b) A series circuit consisting of a 0.1μF capacitor and a 25μH, 20Ω coil is connected across a 10V a.c supply. Calculate the following: **(10 marks)**

i. Resonant frequency

ii. Supply current at resonance

iii. Voltage across the capacitor at resonance

12. With the aid of a labelled diagram, show the construction and describe the operation of a moving coil permanent magnet type instrument **(13 marks)**

13. a) Explain the term ideal transformer **(3 marks)**

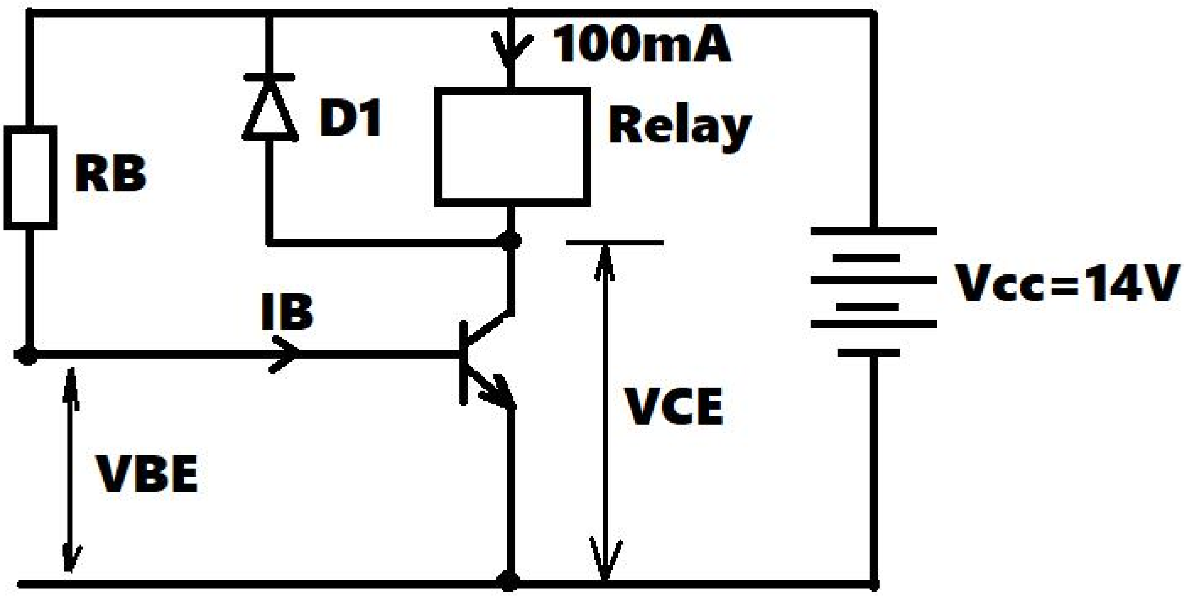
b) A transformer which has a primary winding resistance of 1Ω and a secondary winding resistance of 0.1Ω is connected to a 10Ω load. if the primary current is 0.5A and secondary is 5A, calculate: **(6 marks)**

i. Load power

ii. Copper losses

c) If the transformer in (b) has a core loss of 5W, calculate the efficiency of the transformer **(4 marks)**

14. The transistor in the circuit shown in **Figure 2** below is biased to supply the relay with 100mA. the dc current gain, β, of the transistor is 50 and the forward base-emitter voltage, V_{BE} is 0.7V



a. If the voltage drop across the relay is 12V, calculate the: **(11 marks)**

i. Base current I_B

ii. Emitter current I_E

iii. Voltage between collector and emitter V_{CE}

iv. The base resistance R_B

b. Explain the function of diode D1 in the circuit **(2 marks)**

15. Figure 3 shows three views of an object drawn in first angle orthographic projection. make isometric drawing of the object taking A as the closest point (13 marks)

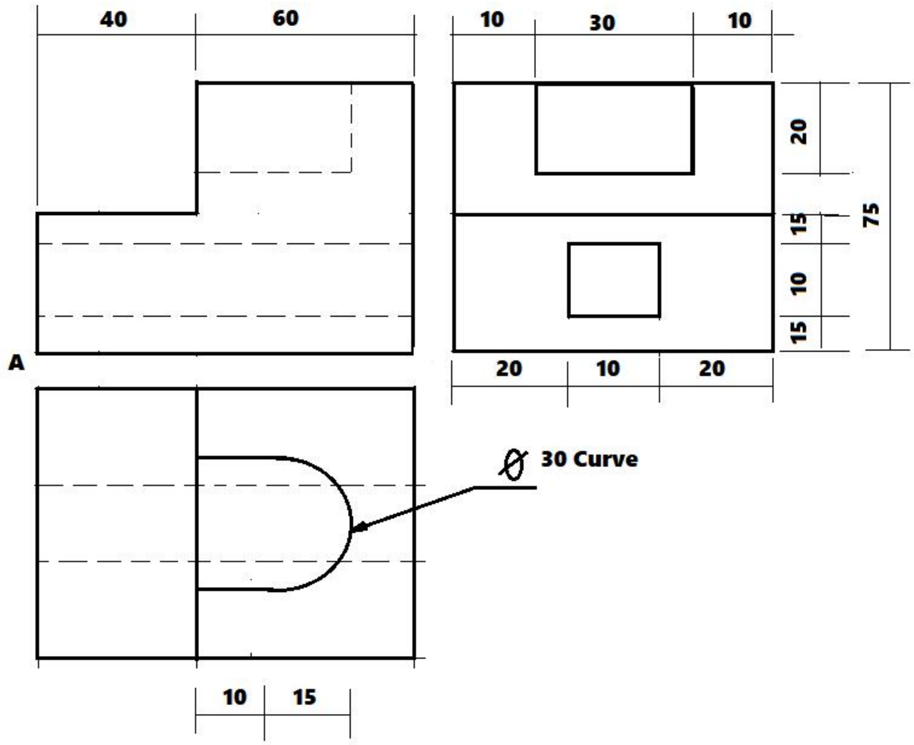


Figure 3